

PUNTA CANA INTL, Dominican RepublicWMO: **784790**Lat: **18.567N**Lon: **68.363W**Elev: **47**StdP: **14.67**Time Zone: **-4.00 (W04)**Period: **96-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	67.6	69.4	61.2	80.9	74.6	63.0	86.2	74.9	20.4	77.4	18.9	77.5	1.0	0	0.403

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.5	89.7	80.0	88.9	79.7	88.1	79.5	81.3	87.1	80.7	86.8	80.1	86.1	11.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.2	152.0	85.1	79.0	150.9	85.0	78.7	149.0	84.7	45.0	88.3	44.5	87.0	43.9	86.5	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.8	17.9	16.5	62.1	91.2	2.7	2.0	60.1	92.7	58.6	93.9	57.0	95.0	55.1	96.5		
			61.2	82.6	2.7	0.9	59.2	83.2	57.7	83.8	56.2	84.3	54.3	84.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	80.0	77.1	77.4	77.5	79.2	80.2	81.7	82.9	82.9	82.2	81.3	79.6	78.4	(6)
(7)		DBStd	2.74	1.99	1.84	2.01	1.72	1.72	1.54	1.62	1.74	1.67	1.78	1.86	2.09	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD50	10966	840	767	854	875	936	952	1019	1019	966	971	887	880	(10)
(11)		CDD65	5491	375	347	389	425	471	502	554	554	516	506	437	415	(11)
(12)		CDH74	53564	2697	2462	2878	3699	4608	5508	6491	6555	5861	5320	4038	3447	(12)
(13)	CDH80	15082	344	339	489	778	1171	1746	2321	2452	2140	1756	976	570	(13)	
(14)	Wind	WSAvg	8.5	8.9	9.3	9.2	9.2	8.1	7.9	9.5	9.1	7.1	6.5	7.5	9.5	(14)
(15)	Precipitation	PrecAvg	42.2	2.6	2.1	2.2	2.8	4.8	4.2	3.1	3.7	3.9	5.7	4.1	3.1	(15)
(16)		PrecMax	65.4	6.0	6.9	6.2	8.7	13.7	8.2	8.1	10.7	9.9	22.2	10.9	7.2	(16)
(17)		PrecMin	25.9	0.5	0.2	0.4	0.0	0.3	0.4	0.9	0.9	0.3	1.4	0.8	0.6	(17)
(18)		PrecStd	9.8	1.3	1.5	1.5	2.2	3.4	2.1	1.5	2.3	2.2	4.2	2.6	1.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.5	85.8	86.3	87.5	88.2	89.5	89.9	90.0	90.0	89.8	87.7	86.1	(19)
(20)			MCWB	75.2	75.7	76.4	78.7	79.6	80.2	80.0	80.2	80.0	79.6	79.3	76.6	(20)
(21)		2%	DB	83.6	84.0	84.6	86.0	87.3	88.0	89.2	89.5	89.6	88.2	86.3	84.5	(21)
(22)			MCWB	74.8	74.5	75.0	77.2	78.9	79.6	79.8	80.0	79.8	79.4	78.2	76.1	(22)
(23)		5%	DB	82.6	82.7	83.9	84.6	86.0	87.5	88.0	88.2	88.2	87.8	85.8	84.0	(23)
(24)			MCWB	74.5	73.9	74.4	75.8	78.0	79.3	79.3	79.7	79.7	79.3	77.8	75.9	(24)
(25)		10%	DB	82.2	82.3	82.6	84.0	84.6	86.2	87.6	87.8	87.7	86.4	84.5	82.7	(25)
(26)			MCWB	74.3	73.8	73.5	75.3	77.1	78.6	79.1	79.5	79.6	78.7	77.0	75.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.0	77.6	77.6	80.1	81.6	82.0	81.8	81.8	81.9	81.1	80.1	78.7	(27)
(28)			MCD B	81.7	82.4	84.4	86.0	86.9	87.1	87.8	87.7	87.3	86.8	85.1	83.0	(28)
(29)		2%	WB	76.5	76.2	76.3	78.5	80.0	80.6	80.6	80.9	80.9	80.5	79.5	77.7	(29)
(30)			MCD B	81.5	81.7	82.6	83.8	85.3	86.3	87.0	87.2	87.0	86.2	84.5	82.6	(30)
(31)		5%	WB	75.6	75.3	75.4	77.5	78.9	79.9	80.0	80.2	80.3	79.9	78.7	76.9	(31)
(32)			MCD B	81.0	81.1	81.6	82.8	84.2	85.5	86.1	86.5	86.4	85.6	83.8	82.1	(32)
(33)		10%	WB	74.9	74.5	74.6	76.4	78.1	79.3	79.6	79.9	79.9	79.3	78.0	76.2	(33)
(34)			MCD B	80.3	80.3	80.7	81.8	83.3	84.8	85.5	85.9	85.8	85.0	83.0	81.5	(34)
(35)	Mean Daily Temperature Range		MDBR	10.7	10.5	10.8	10.4	10.4	10.3	10.0	10.5	11.6	11.8	10.7	10.0	(35)
(36)		5% DB	MCD BR	11.1	11.0	12.0	10.9	11.2	11.0	10.4	10.8	12.4	12.7	11.4	10.6	(36)
(37)			MCWBR	6.1	5.7	5.8	5.3	5.5	5.3	4.8	5.2	6.1	6.2	6.0	5.5	(37)
(38)		5% WB	MCD BR	10.7	10.7	11.3	10.8	10.8	10.6	10.0	10.5	11.8	12.0	10.6	9.9	(38)
(39)			MCWBR	6.5	6.1	6.0	6.0	6.0	5.9	5.1	5.4	6.2	6.3	6.0	5.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.366	0.380	0.383	0.410	0.441	0.513	0.507	0.482	0.428	0.387	0.388	0.381	(40)	
(41)		taud	2.568	2.521	2.506	2.461	2.394	2.139	2.158	2.255	2.445	2.567	2.559	2.548	(41)	
(42)		Ebn at Noon	291	292	294	285	272	251	253	261	276	286	282	283	(42)	
(43)		Edh at Noon	30	33	35	36	39	49	49	44	36	31	30	30	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1460	1700	1887	1979	1915	1947	2008	1939	1808	1616	1430	1371	(44)	
(45)		RadStd	57	46	69	111	121	94	86	76	122	117	106	62	(45)	